

Automatic Feature Selection Based Motor Imagery Movements Detection Scheme from EEG Signals in the Dual Tree Complex Wavelet Transform Domain

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Abstract—An automatic feature selection based classification scheme in the Dual Tree Complex Wavelet Transform (DTCWT) domain from electroencephalogram (EEG) signals has been presented in this study to identify left and right hand imagery movements. First, the EEG epochs are decomposed into different real and imaginary coefficient bands and then, some statistical features like norm entropy and mean absolute deviation (MAD) have been calculated. These features are combined into a single feature space and after that optimal features have been selected automatically imposing some feature selection criteria from this combined feature space. The selected features have been justified as suitable to classify different kinds of motor imagery EEG signals by statistical hypothesis testing (e.g., one way ANOVA) and graphical analyses (e.g., scatter plots). Finally, K-nearest neighbor (kNN) based classifiers are developed using these selected features for classifying 2 types of imagery hand movements. 90.36% overall accuracy is achieved in publicly available BCI competition II Graz data set which is shown to be superior than several existing methods.

Index Terms—Electroencephalogram (EEG); Entropy; Dual tree complex wavelet transform; Mean absolute deviation; kNN classifier.

I. INTRODUCTION

Brain computer interfacing (BCI) refers to a computer aided system which is operated and controlled by only intent. The primary purpose of BCI is to assist disable people for their rehabilitation. BCI includes analysis, detection and identification of various kinds of motor imagery movements for real time implementation and control. Electroencephalogram (EEG) signals are the most common in BCI since it can be obtained as non-invasive system [1].

Several categories of EEG-based BCI exist including limb motor imagery classification [2], continuous arm movements direction detection [3] and individual finger movement decoding [4] etc. Various techniques have been proposed in the literature for detecting several kinds of arm movements [5], which is a major category of BCI. A wavelet-based common spatial pattern algorithm using low frequency features and Fisher linear discriminant classifier (LDA) is developed to classify fast and slow hand movements in [6]. In [7], filter bank common spatial pattern is implemented with mutual information based feature selection and for classification, Naive Bayesian Parzen window classifier is used. Wrist movement classification has been done by extracting gamma band features from wavelet packet transform and employing radial basis function classifier in [8]. Eigen-vectors of co-variance matrix along with principle component analysis for dimension reduction and support vector machine (SVM) for classification

is described in [9] for right and left hand wrist imaginary movements detection. Adaptive auto regressive parameters based EEG signals separation method is proposed in [10]. In [11], power spectral density, wavelet coefficients and average power obtained from EEG segments are extracted as features and then different classifiers such as LDA, SVM and kNN are used to detect left and right hand movements. Wavelet packet entropy of EEG signals is used as feature in [12] and after that, SVM is employed as the pattern classifier.

For experimental and research purposes, EEG data can be acquired with different setup and paradigms. However, for evaluating and comparing different algorithms, standard data sets provided by BCI competition are extensively used. Among several available BCI data sets, BCI competition II Graz motor imagery data set has been widely studied to identify imagery hand movements by different approaches. Power spectral density and linear discriminant analysis (LDA) based EEG signal classification is described in [13]. In [14], Hidden Markov Model (HMM) based method is presented by the same author. In [15], adaptive auto regressive (AAR) model based features along with multi layer perceptron and Bayesian Graphical Network are stated. For extracting features from mu rhythms, Morlet wavelet is used in [16] and for classification Bayes quadratic classifiers are deployed.

The goal of this study is to classify imagery hand movements by automatically extracted features from EEG signals in the dual tree complex wavelet transform (DTCWT) domain. DTCWT is very popular for image and video processing ([17], [18]), and recently, it's applications are found in bio-medical signal processing ([19], [20]). Since motor movements occur in the low frequency EEG bands [21], to analyze specific bands extensively, DTCWT has been used since DTCWT is a rich level analysis tool than traditional discrete wavelet transform (DWT). The product of norm entropy and signal mean absolute deviation obtained from the various DTCWT bands are used as features. The ability of this value in discriminating imagery hand movements is demonstrated using various types of scatter plots, box plots, one way ANOVA and Kruskal-Wallis analyses. An automatic selection method based on J_3 criteria is proposed for selecting the optimum feature vector. Employing a number of classifiers, left and right hand imagery movements are classified with these features. The performance of these classifiers in detecting hand movements obtained and compared with those of several existing techniques.

II. EEG DATABASE DESCRIPTION

In this paper, we have used BCI competition II Graz motor imagery data set which is recorded by Technical University of Graz. The data was acquired when the normal subject was trying to control a bar (feedback bar) while sitting in a chair with armrests and looking at the screen. For controlling the bar position, the subject made left or right hand imagery movements. The movement instructions were given in random order [22].

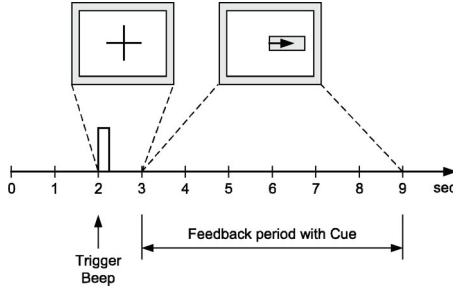


Fig. 1. Timing scheme of the experiment.

The experiment consists of 7 runs with 40 trials each. At 2 second of each trial, a cross “+” was displayed for 1 s. After this, an arrow (either left or right) was displayed, which was the cue. The subject was instructed to move the bar into the cue direction and the movement was controlled by adaptive auto-regressive parameters, recorded from both C3 and C4 channels. The EEG signal was band-pass filtered between 0.5 and 30 Hz and for the experiment, the sampling frequency was 128 Hz. More detail description of the EEG recording set up is described in [23]. Fig. 1 (a) illustrates the timing scheme of the EEG experimental paradigm.

III. DUAL TREE COMPLEX WAVELET TRANSFORM

A recent enhancement of Discrete Wavelet Transform (DWT) is Dual tree complex wavelet transform (DTCWT). It has some additional properties including directionally selective in two and higher dimensions and shift invariant[24]. For any d dimensional signal, DTCWT is 2^d times redundant when compared to DWT. Hence it offers directional information in six directions. As a result, it is more effective in localizing EEG signals both in time and frequency. DTCWT deploys two real DWTs; the first one gives the real part of the complex transform whereas, the second DWT represents the imaginary part. The filter bank structure to implement DTCWT is given in Fig. 2.

If square matrices $\mathbf{H}_h$ and $\mathbf{H}_g$ indicate the two real DWTs, then the DTCWT can be expressed as follows:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_h \\ \mathbf{H}_g \end{bmatrix} \quad (1)$$

The inverse transform of $\mathbf{H}$ is defined as,

$$\mathbf{H}^{-1} = \frac{1}{2} \begin{bmatrix} \mathbf{H}_h^{-1} & \mathbf{H}_g^{-1} \end{bmatrix} \quad (2)$$

Now, if a real signal is represented by the vector $\mathbf{x}$, then $\mathbf{w}_h = \mathbf{H}_h \mathbf{x}$ denotes the real part of the DTCWT whereas $\mathbf{w}_g = \mathbf{H}_g \mathbf{x}$ is the imaginary part.

When a real signal is decomposed using DTCWT, the upper filter banks give the real part while the lower filter

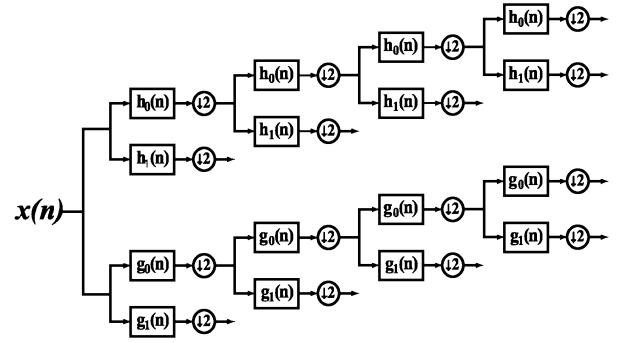


Fig. 2. Analysis filter bank structure of DTCWT.

banks produce the imaginary parts of the complex coefficients. Among various available wavelets, Farras wavelets [25] are used here. The EEG signals obtained from C3 channel during an arbitrary trial for left and right hand imagery movements and their corresponding second level DTCWT coefficients are shown in Fig. 3.

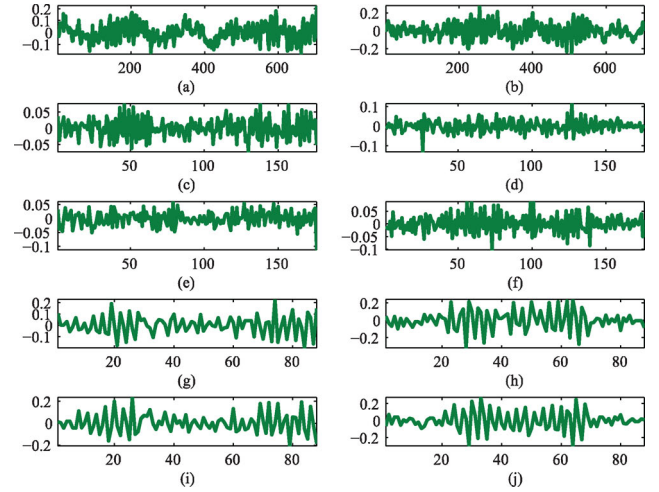


Fig. 3. (a)-(b) EEG signals obtained from C3 electrode for left and right hand imagery movements, respectively; (c)-(d) real coef. band of level 1 for left and right movements, respectively; (e)-(f) imaginary coef. band of level 1 for left and right; (g)-(h) real coef. band of level 2 for left and right; (i)-(j) imaginary coef. band of level 2 for left and right hand imagery movements, respectively.

IV. DUAL TREE COMPLEX WAVELET TRANSFORM DOMAIN BASED ANALYSIS

Our proposed method consists of e parts- first is the decomposition of the EEG signals of each trial with DTCWT into 3 levels and suitable feature extraction. Second, optimal features are selected automatically using J_3 criteria. And finally, for classification purpose, a patter classifier is deployed (e.g., kNN classifier).

A. DTCWT Domain Based Feature Extraction

In the Graz imagery data base experiment paradigm, feedback was taken from C3 and C4 channels and Cz electrode was the reference. As a result, for a trial epoch, DTCWT is applied to decompose EEG signals (both C3 and C4) into 3 levels. If the original epoch is expressed as X which contains 0.5 to 30 Hz frequency components, after three

levels of DTCWT, the output bands are $Y_1(16-30 \text{ Hz})$, $Y_2(7.5-15 \text{ Hz})$, $Y_3(3.75-7.5 \text{ Hz})$ and $Z_3(0.5-3.75 \text{ Hz})$. Reconstructions of these bands using the inverse DTCWT approximately leads to the physiological EEG sub-bands (e.g., delta, theta, alpha and beta, respectively) [26]. Now, all coefficient bands have both real and imaginary parts and as a result, after 3 level decomposition total 4 real and 4 imaginary bands are created for each channel. From these on, in this work RB_x and IB_x will be used to denote the real and imaginary parts and the corresponding levels where 'x' is the level indicator.

In the frequency domain, the dynamics behind EEG signals is spread over various sub-bands. We need to extract information in lower frequency bands of EEG signals and μ rhythms [16] for analyzing imagery motor movements. For this purpose, mean absolute deviation and norm entropy from different frequency bands have been extracted as features.

1) *Mean Absolute Deviation (MAD)*: Mean Absolute Deviation (MAD) of a data set is defined as the average distance between each data value and the mean value; thus providing a measure of statistical dispersion-

$$MAD = \frac{1}{N} \sum_{i=1}^N |x_i - \bar{x}| \quad (3)$$

Here N denotes number of data points, x_i is the i -th sample and $\bar{x}$ is the mean of N samples.

2) *Norm Entropy*: Entropy is a common concept in many fields, mainly in signal processing. The entropy E must be an additive cost function such that $E(0) = 0$ and $E(s) = \sum E(s_i)$, where S is the signal and S_j are the coefficients of $\hat{S}$ in an orthonormal basis.

The concentration in l^p norm entropy [32] is given as:

$$E(s_j) = |S_j|^p, \quad S_0, \quad E(s) = \sum_i |S_i|^p = \|S\|_p^p, \quad 1 \leq p \quad (4)$$

However, since MAD and norm entropy both do not have good p-values individually in one way analysis of variance (ANOVA), they do not possess the ideal feature quality separately to discriminate left and right hand imagery movements. Also by using two features from each band, a total of $4 \times 2 \times 2 \times 2 = 32$ features are needed to be used for four levels from two channels which has very large search space for optimal feature selection. To overcome these problems, we have defined a combined feature space as the product of entropy and MAD. This new feature space has two advantages. One is the separability of this combined feature space between two classes is better than the individuals (very small p-values of ANOVA and Kruskal-Wallis analyses ensure this). The second advantage is that the length of feature vector is now reduced to half compared to separately using entropy and MAD and thus keeping the search space smaller. So, the proposed method extracts features from norm entropy $\times$ MAD plane for all four levels (i.e., eight bands) from two channels.

B. Automatic Feature Selection

Unlike the previous DTCWT domain works, where features are used without automatic selection ([27]-[28]), in this work best feature combination is selected automatically. Since the four level DTCWT of the EEG signal leads to eight coefficient bands for each channel, a total of 16 features are available from

two channels. To select the best feature vector, J_3 criteria is used [29]. J_3 is a cost function for class separability measurement defined as

$$J_3 = \text{trace}\{\mathbf{S}_w^{-1} \mathbf{S}_m\} \quad (5)$$

where $\mathbf{S}_w$ is the within class scatter matrix given by

$$\mathbf{S}_w = \sum_{i=1}^M P_i \mathbf{S}_i \quad (6)$$

Here, P_i is the priori probability of a class and $\mathbf{S}_i$ is the corresponding co-variance matrix. The mixture matrix $\mathbf{S}_m$ can be expressed as

$$\mathbf{S}_m = E[(\mathbf{x} - \boldsymbol{\mu}_0)(\mathbf{x} - \boldsymbol{\mu}_0)^T] \quad (7)$$

where $\boldsymbol{\mu}_0$ is global mean vector. Sometimes between class scatter matrix $\mathbf{S}_b$ is used instead of $\mathbf{S}_m$.

J_3 criteria has the advantage that it is invariant under linear transform [30]. For a feature vector with a particular number of features, the combination yielding maximum value of J_3 is selected as the optimal feature vector. Here from available 16 features, 4 optimal features have been chosen automatically. The four features found using J_3 criteria are $RB_3(C3)$, $RB_4(C3)$, $RB_1(C4)$ and $IB_3(C4)$.

TABLE I
P-VALUES OF THE FEATURES.

Analysis Name	Features	p-values
One way ANOVA	$RB_3(C3)$	5.60 e-18
	$RB_4(C3)$	1.48 e-36
	$RB_1(C4)$	1.20 e-03
	$IB_3(C4)$	7.96 e-05
Kruskal-Wallis test	$RB_3(C3)$	1.70 e-19
	$RB_4(C3)$	2.85 e-24
	$RB_1(C4)$	2.91 e-05
	$IB_3(C4)$	1.39 e-06

Table I shows the best possible features for a 4 element feature vector selected using the J_3 criterion and the corresponding statistical hypothesis results. We have used p-values of one way ANOVA and Kruskal-Wallis [31] test here. The p-value hypothesis states that at least one trial (sample) mean is significantly different than the other trial means statistically if $p < 0.05$ [32]. From Table I, it is seen that these four features have very small p-values which statistically indicates they can be used as good features.

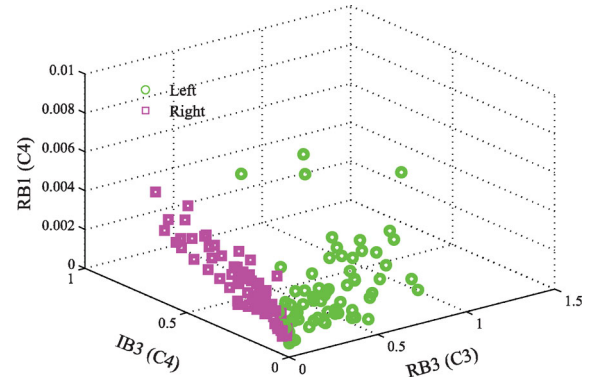


Fig. 4. (a)-(b) Scatter plots of $RB_1(C4)$, $IB_3(C4)$ and $RB_3(C3)$ during 2 different imagery movements (i.e., left and right, respectively).

Apart from ANOVA and Kruskal-Wallis analyses, scatter plots and box plots are provided to further illustrate the classification quality of the selected features. Fig. 4 represents the scatter plots of $RB_1(C4)$, $IB_3(C4)$ and $RB_3(C3)$. In Fig. 4, green circle and magenta square markers indicate the feature values during left and right hand movements, respectively. The markers clearly span different regions, which signifies their variation during the two specified movements.

The box plots of $RB_4(C3)$ and $IB_3(C4)$ for left and right hand imagery movements, respectively are presented in Figures 5-6. The boxes have non overlapping notches for the two specific movements which indicates that the features have sufficiently distinguishable values.

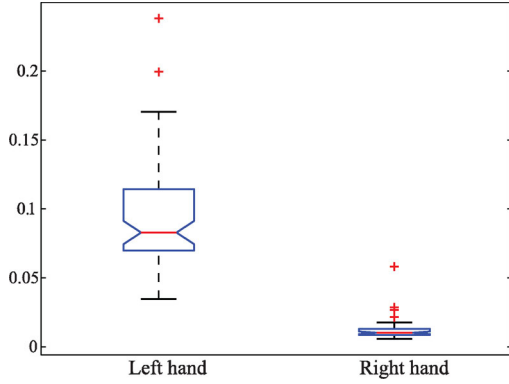


Fig. 5. Boxplot of RB_4 (from C3 channel) for left and right hand movements, respectively.

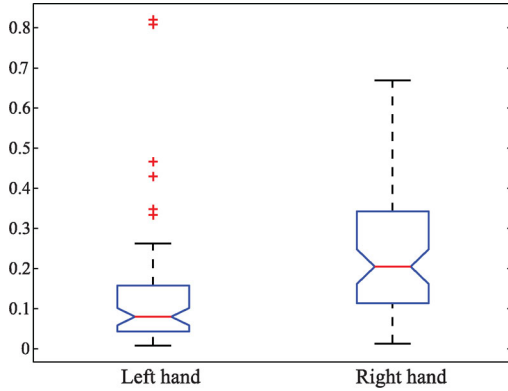


Fig. 6. Boxplot of IB_3 (from C3 channel) for two different imagery hand movements, respectively.

The graphical results (e.g., box plots, scatter plots) and the hypothesis tests (p-values of ANOVA-1 and Kruskal-Wallis) indicate that automatically extracted $RB_3(C3)$, $RB_4(C3)$, $RB_2(C4)$ and $IB_3(C4)$ using J_3 feature selection criteria from the combined feature space formed by MAD and norm entropy have discriminating values for left and right hand imaginary movements. In the feature vector space, these features possess small within-class distance and good between-class distance [30] and as a result they may be used as suitable features for classifying right and left hand imagery movements from EEG epochs.

V. EXPERIMENTAL RESULTS

The Graz data set has 140 trials for left hand imagery and 140 trials for right hand imagery movements. Each trial

contains signals of 9 seconds. For extracting features, EEG signals after the given “cue” is used from both C3 and C4 electrodes. Only four features have been used to form the feature vector (i.e., 4 selected features by J_3 criteria). The train and test feature matrices having 140×4 dimension are used for classification.

The simulation configuration was: Windows-7 32 bit, 1 GB RAM, 2.93 GHz processor (Intel Core 2 Duo) and MATLAB 2013b [32]. Both training and test data set have been using leave one out cross validation method.

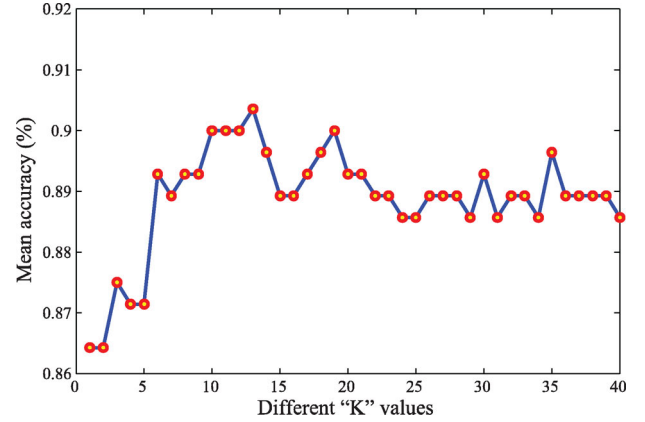


Fig. 7. Variation of accuracy(%) for different “K” values.

First, in Fig. 7, the mean accuracy of the classification employing kNN classifier with different values of “K” is shown and from the figure it is noted that for this identification problem, $K=13$ gives the highest accuracy of 90.36%. The distance parameter of the kNN classifier is selected as “Euclidean” distance [32].

TABLE II
ACCURACY OF VARIOUS CLASSIFIERS

Classifier name	Classifier parameter	Mean Accuracy (%)
Naive Bayes	Normal distribution	84.64
Discriminant Analysis	Mahalanobis distance	85.71
	Diagquadratic	85.71
	Quadratic	86.43
	Linear	86.79
	Diaglinear	87.14
SVM	Radial Basis kernel	87.14
	Linear kernel	88.21
kNN	Cosine distance	88.93
	Cityblock distance	90.36
	Euclidean distance	90.36

Next, Table II shows the performance comparison of different classifiers with varying parameters indicating that kNN classifier with “Euclidean” and “Cityblock” distance gives better accuracy than others.

Lastly, the accuracy of the proposed method has been compared in Table III with several existing methods. From Table III, it can be stated that by deploying kNN classifier, our method has been able to detect left and right hand imagery movements with a better accuracy than other listed methods.

VI. CONCLUSION

An automated method has been developed in this work to classify right and left hand imagery movements offering

TABLE III
MEAN ACCURACY COMPARISON

Method	Proposed By	Classifier	Mean Accuracy (%)
PSD	Solhjoo et al. [13]	Mahalanobis distance	63.1
		Gaussian classifiers	65.4
		LDA	65.6
Raw EEG	Solhjoo et al. [14]	HMM	77.5
AAR	Tavakolian et al. [15]	Bayes Quadratic	82.86
		BGN	83.57
		MLP	84.29
Morlet wavelet	Lemm et al. [16]	Bayes Quadratic	89.29
Proposed method (DTCWT)		kNN	90.36

promising supports for brain computer interfacing. By extracting features from a combined space formed by the product of mean absolute deviation and norm entropy of the EEG signals in the DTCWT domain, EEG epochs have been successfully classified. Optimal combination of features have been selected automatically using J_3 criteria and justification of this feature combination has been provided with scatter plots, box plots, one way ANOVA analysis as well as Kruskal-Wallis test. Among different types of classifiers including SVM, kNN, Bayes classifier and LDA, kNN has the highest accuracy of 90.36%. Finally, the proposed method has been shown to be better than several existing methods found in the EEG based BCI literature.

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